

InvAct: View and Scene-invariant Atomic Action Learning from Videos

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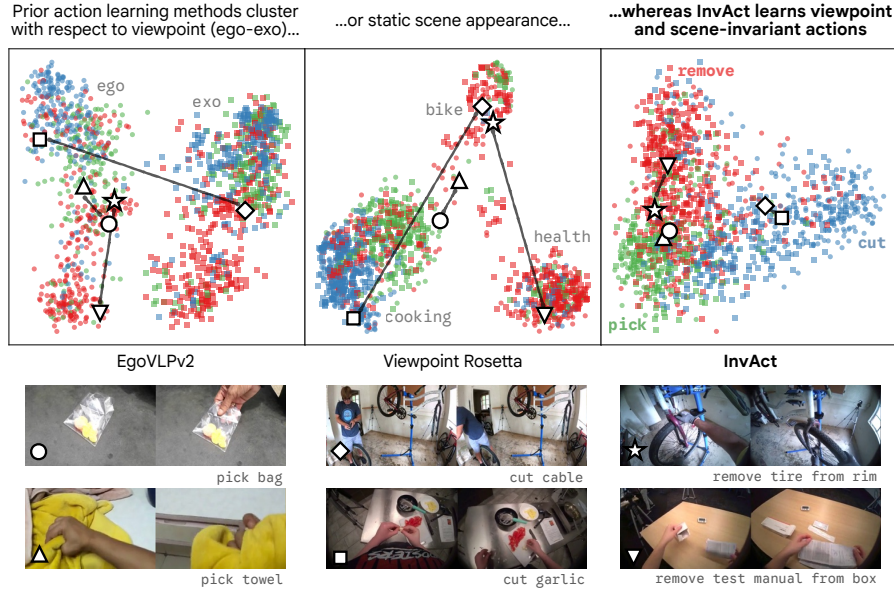


Figure 1: In the top row, colors indicate action semantics (e.g., pick, cut, remove), while colored circles and squares denote ego-view and exo-view embeddings, respectively. Hollow markers (circle, star, triangle, etc.) correspond to the example videos shown in the bottom row. The desired representation should group together the same actions across different viewpoints and scenes. EgoVLPv2 [38] clusters video action embeddings by viewpoint (ego-exo), whereas Rosetta [31] clusters by scene (bike, cooking, health). For instance, two cut samples sharing the same dynamics, cut cable (bike) and cut garlic (cooking), end up far apart in EgoVLPv2 and Viewpoint Rosetta due to viewpoint and scene mismatch, respectively. Conversely, two samples sharing the same scene (bike) but different dynamics, cut cable and remove tire, are placed close together by Viewpoint Rosetta despite being different actions. **InvAct** clusters embeddings based on fine-grained action semantics, regardless of scene or viewpoint. As illustrated, samples depicting the same action are connected by lines, emphasizing that InvAct successfully aligns action representations across diverse conditions.

Abstract

Action learning from videos should learn to produce representations that faithfully reflect similarity of individual actions performed in the scene. Yet, recent methods often overfit to scene or viewpoint cues, yielding clusters dominated by spurious appearance or viewpoint similarity rather than action similarity. We address this with **InvAct, the first atomic action embedding model invariant to both ego-exo viewpoint shifts and scene changes**. To reduce reliance on scene cues, we propose *structured token commitment*, a hard row-wise attention mask that routes each video token to a single special-token slot. Action-relevant tokens align with the class token, while non-action tokens align with register tokens that are discarded at

inference. We further introduce a *unified scene-aware contrastive objective* that attracts same-action clips across scenes and repels different actions within the same scene. InvAct significantly outperforms baselines on cross-view and cross-scene action retrieval, transfers strongly to unseen datasets, reduces scene bias, composes atomic embeddings into long-action retrieval, and shows promising results on robotic manipulation. To inspect InvAct’s embeddings, see the interactive demo: https://three-bee.github.io/invact_website/

1 Introduction

Action representation learning aims to produce representations that support downstream tasks such as action recognition [11, 5], retrieval [31], skill transfer [47], imitation learning [24, 46], vision-language-action model (VLA) pretraining for robotics [9], and conditioning world models [16]. An ideal action embedding should be invariant to changes in viewpoint, scene, and object appearance.

While invariance is a key goal, it is important to note that actions vary in granularity. **Keystep actions** (e.g., *checking for damages*, *cooking an omelet*) describe longer, higher-level procedures that correlate strongly with objects, scene context, and appearance [19]. These can be decomposed into **atomic actions** (e.g., *pushing*, *pulling*, *cutting*), which are short, interaction-centric primitives.

To advance action learning, some recent works [49, 19, 37] have focused on **view-invariance** at the **keystep** level, aligning representations across egocentric and exocentric viewpoints using large-scale datasets such as Ego-Exo4D [19]. However, when view-invariance becomes the primary objective, models risk relying on **static scene cues** rather than genuine action dynamics (see Fig. 1 for an example of resulting clusters). Strong quantitative results may therefore not reflect true action understanding, particularly on low scene-diversity benchmarks such as the cooking subset of Ego-Exo4D, where models tend to fail when the same action occurs across visually distinct scenes.

Fig. 2 illustrates this issue by visualizing retrieval behavior on the Ego-Exo4D validation set, with all scenes included in the retrieval pool. It reveals that recent baselines are over-adapted to appearance cues, ranking top-k clips by shared scene context such as layout and objects rather than interaction dynamics. This failure mode is most evident in the ego→exo examples retrieved by Viewpoint Rosetta Stone [31]: the method returns clips from the same kitchen by matching static cues such as the blue bowl and brown cabinets, missing stronger dynamic matches in different scenes.

To overcome these limitations, we present **InvAct**, the **first atomic-action embedding model** that maps short video clips into a compact representation space **invariant to both scene changes and ego→exo viewpoint shifts**, while remaining discriminative for fine-grained interaction dynamics. Unlike prior work that focuses on viewpoint alignment at the keystep level, InvAct explicitly targets

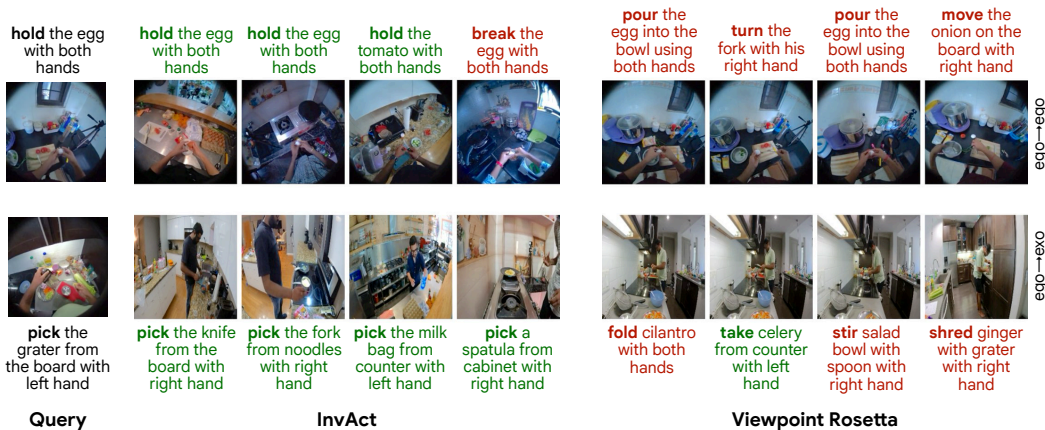


Figure 2: We show the top 4 nearest neighbors retrieved based on the learned latent action space of different methods. Previous baseline rank mostly based on static scene context and the nearest retrievals consist of incorrect, different actions in the same environment. Meanwhile, our method successfully ranks based on action similarity, regardless of the viewpoint (ego/exo) or scene differences. Text descriptions are shown only for visualization and are not used during inference.

both viewpoint and scene invariance for atomic actions, and shows that the resulting embeddings can be composed to support keystone action retrieval across all ego–exo combinations.

Technically, we introduce two novel components. First, **structured token commitment**, a parameter-free mechanism that turns register (REG) tokens from passive auxiliary workspaces into active discard filters. By restricting each video token to a single special-token read source via hard row-wise masking, and combining this with action losses applied only to CLS, we induce a partition in which action-relevant tokens commit to CLS and non-action tokens commit to discardable REG slots. Second, **a unified scene-aware contrastive loss** that simultaneously attracts same-action clips across scenes and repels different actions from the same scene, directly counteracting the scene-based shortcuts of view-invariant prior works [31, 37]. We further leverage sentence- and verb-level action labels to semantically anchor the embeddings, requiring no text at inference time, unlike prior work [31, 38, 27]. Architecturally, we train a transformer-based encoder on large-scale data [19] that tokenizes and fuses optical flow [44], providing scene-invariant motion cues, with pretrained feature maps [43], providing viewpoint-invariant appearance representations, into a single atomic action.

The effectiveness of InvAct is shown in Fig. 2 and Section 4, where it retrieves stable, faithful matches in complex and diverse scenes. Our contributions are as follows:

- To our knowledge, InvAct is the **first atomic-action learning model to tackle both the large viewpoint-invariance and scene-invariance in a single model**.
- We propose **structured token commitment**: a parameter-free, hard attention re-routing that constrains each video token to read only one special-token, [CLS, REG]. (Section 3.3). To our knowledge, routing information to [REG, CLS] with such biasing had not been explored before, either in general or in the context of invariant representation learning.
- We conduct extensive ablations on inputs, learning paradigms, contrastive loss components, and scene-bias properties (Table 3), a missing aspect in literature. Particularly, we demonstrate the importance of **our novel unified scene-aware contrastive loss**, enforcing cross-scene action attraction and same-scene action repulsion, a combination absent from prior work (Section 3.2).
- We show that long (keystone) actions can be composed from our short (atomic) embeddings without additional training, enabling out-of-the-box long-video action retrieval (Section 4.2 and Table 1).
- We outperform a diverse set of prior methods, spanning general purpose [39, 43, 4, 1, 5], robot-centric [50, 22] and human-centric [18, 51, 27, 38, 45, 49, 37, 19, 31] action learning methods on video-to-video retrieval for both atomic and keystone actions in most cases, without any keystone-level training (Table 1). We further test our method on four additional datasets [42, 17, 29, 23] without any finetuning, demonstrating strong zero-shot generalization capability (Table 2).
- We validate our model through long-sequence probing, latent clustering, attention map, scene-bias, and linguistic alignment analyses. Specifically, we demonstrate a VLA pretraining task using InvAct’s embeddings, outperforming robot-centric action learning baselines [50] (Section 4.2).

We believe InvAct and our findings can benefit both robotics and human action understanding, and help bridge robot- and human-oriented action learning. We will release code and trained models.

2 Related Work

Generalizable representations. Large pretrained models like CLIP [39], DINOv3 [43], and TimeSFormer-based architectures [8] provide strong representations of large-scale training. V-JEPA 2 [4] further improves temporal modeling of videos, while FlowFeat [1] emphasizes motion-driven features. LiFT [5] converts image features into time-sensitive video descriptors by linearizing feature trajectories. However, none of these models are designed to be scene and view-invariant.

Multimodality for action learning. Video–language pretraining (VLP) methods such as LaViLa [51] and X-CLIP [32] focus on improving video–text alignment using language supervision and multi-grained contrastive matching between frame-text. HierVL [3] complements this by learning hierarchical video–text embeddings that align both clip-level narrations and long-video summaries. Egocentric VLP methods like EgoVLPv2 [27, 38] adapt video–text contrastive learning to first-person data, by mining egocentric-aware positives/negatives and using stronger cross-modal fusion. VL-JEPA [12] explores predicting text embeddings from videos rather than generating visual tokens.

TARA [6] adapts multimodal LLMs to time-sensitive video-text embeddings with temporal order. These directions complement ours by improving semantic grounding through video-text alignment, similar to how we use atomic labels. However, they do not explicitly enforce scene-view invariance at the atomic level, and they often inject text into the model, whereas we use language through loss.

Viewpoint in action learning. Ego-Exo4D benchmark formalizes synchronized first- and third-person understanding on scale [19], and works like Viewpoint Rosetta Stone [31] pursue view-invariant alignment in weakly paired settings. SUM-L [45] focuses on egocentric learning, but utilizes unpaired ego-exo videos with text label guidance. AE2 [49] learns fine-grained invariance by temporally aligning ego-exo videos without synchronization, BYOV [37] uses masked ego-exo modeling to encourage cross-view consistency. Differently, Seeing Without Pixels [48] reveals that actions can be observed through non-appearance signals like camera motion, aligning trajectory cues with language. In contrast, we are the first to tackle scene invariance while still satisfying viewpoint invariance. We show the benefit our method using all subsets of the Ego-Exo4D validation split, going beyond the more constrained subsets or settings used in prior work [45, 49, 31, 37].

Latent atomic action discovery in robotics. LAPA [50], LAOF [10], and LAPO [40] study latent action discovery in embodied settings, learning action abstractions from video with minimal or no action labels. LAOF also uses optical-flow constraints to stabilize learning. VILA [22] focuses on supervised view-invariant latent actions for policy learning. DreamDojo [16] uses latent actions in physics-aware world model pretraining and conditioning. Ko *et al.* [26] extract action-relevant structure from videos via flow and depth for policy learning. These works share our intuition that motion aids in action learning when appearance varies. However, we learn atomic actions that stay consistent across viewpoint shifts with complex environments and grounded by language, a setting that is not commonly studied in such robotics-oriented work.

Special tokens and attention re-routing. We consider CLS and REG (register) tokens as special tokens. Register tokens [14] provide auxiliary workspace in vision transformers, absorbing internal computation while being discarded at the output. Prior works have explored structured attention routing in various forms: Slot Attention [30] used token-to-slot assignment for object-centric decomposition; Mask2Former [13] applied masked cross-attention to restrict queries to predicted mask regions; WSSS [20] introduced class-specific token masking for weakly supervised segmentation; and Marouani *et al.* [33] showed that specialized CLS-patch processing paths improve separation between global and local computations. However, none of these works explicitly re-route information between special tokens, and not in invariant representation learning.

3 Method

We propose to learn a coherent, **view- and scene-invariant atomic action space** from video sequences, yielding embeddings that generalize across viewpoints and environments without relying on scene-based shortcuts. We show that a transformer, trained with our contrastive and classification losses combined with our attention re-routing technique, can achieve this challenging task.

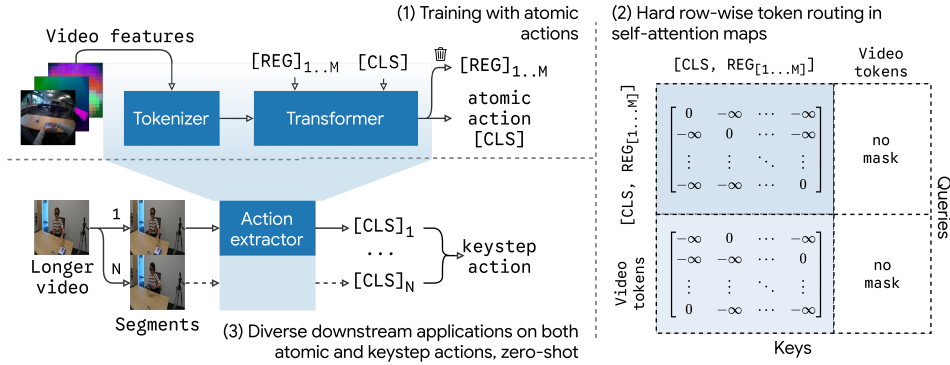


Figure 3: Architecture of our action extractor. We train a foundational backbone with large-scale atomic video sequences (1) with contrastive and semantic losses, and attention routing (2). This yields high performance on diverse downstream applications such as view and scene-invariant action retrieval, action classification, and VLA pretraining, in both short- and long-level actions (3).

3.1 Architecture

The architecture is illustrated in Fig. 3. Given an atomic video clip, we extract two complementary streams: optical flow [44], which captures scene-invariant motion dynamics, and frozen pretrained frame features [43], which provide viewpoint-robust appearance representations. These streams are fused within a transformer encoder to produce a single atomic action token (CLS). Flow tokens are tokenized via our trained tokenizer [2] and passed to the transformer alongside REG tokens, while frozen video features are injected through cross-attention. Within the transformer, our additive masking over the self-attention map constrains each video token to read special-token context from a single slot, inducing a learned partition where non-action tokens commit to REG slots that are discarded at inference. For atomic-level clips, a single CLS token serves as the action embedding, while for keystone-level videos, atomic chunk embeddings are chained to form the action embedding. Input justifications and their effects on scene-view invariance are discussed in Appendix D.

3.2 Unified scene-aware contrastive loss

We build on the contrastive losses of prior work [31] and introduce two main loss additions: a unified scene-aware loss with attraction-repulsion terms, and atomic-level semantic guidance. These are designed to reduce scene bias and disentangle scene and viewpoint information from the learned representations. We elaborate here only on the visual contrastive pairs; the full loss formulation and ablations are provided in Appendix F.2.

We define a **novel unified scene-aware contrastive loss** over the batch. For each anchor i and atomic action embedding $v = \text{CLS}$, let $\mathcal{P}(i)$ denote clips sharing the **same action from different scenes**, $\mathcal{R}(i)$ denote clips from the **same scene with a different action**, and $\mathcal{N}(i)$ denote clips with a **different action from a different scene**. The scene-aware contrastive loss is given in Eq. (1):

$$\mathcal{L}_{\text{scene}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\sum_{p \in \mathcal{P}(i)} \exp\langle v_i, v_p \rangle}{\sum_{p \in \mathcal{P}(i)} \exp\langle v_i, v_p \rangle + \sum_{r \in \mathcal{R}(i)} \exp\langle v_i, v_r \rangle + \sum_{n \in \mathcal{N}(i)} \exp\langle v_i, v_n \rangle}. \quad (1)$$

Eq. (1) attracts cross-scene same-action pairs via $\mathcal{P}(i)$, and repels same-scene different-action pairs via $\mathcal{R}(i)$ and cross-scene different-action pairs via $\mathcal{N}(i)$. Together, these terms discourage appearance-based shortcuts and encourage a richer atomic action space. The objective can further be decomposed into attract-repel and ego-exo components, as detailed in Appendix F.2.

Notably, prior methods like Viewpoint Rosetta [31] omit same-scene repulsion, since scene bias helps keystone-level viewpoint invariance, but fails on fine-grained atomic actions, observed in Fig. 2 and Table 1. InvAct’s additions instead improve both cross-scene and cross-view atomic action retrieval (Table 3, losses) while remaining competitive on keystone action retrieval, even in zero-shot settings by only combining its atomic action embeddings (Table 1).

3.3 Structured token commitment

Actions in video clips are typically confined to a small spatial region, such as a hand contacting an object, while most tokens carry background or non-action content. A standard soft attention mixture lets every video token feed a portion of its content into CLS, regardless of whether it participates in the action. This leakage opens the door to scene-based shortcuts (Fig. 2), where CLS ends up encoding static scene information alongside or instead of the action itself. To prevent this leakage and keep CLS built from action-relevant content, we introduce **hard token re-routing**.

Mask design. We achieve this via a mask M (block (2) in Fig. 3) applied to the self-attention logits as $(QK^\top + M)$ before softmax, where Q and K are query and key matrices, respectively. Let $\mathcal{S}$ and $\mathcal{X}$ denote the sets of special [CLS, REG] and video token indices, respectively. The mask enforces two routing rules: first, CLS and REG are blocked from attending to each other, in Eq. (2):

$$M_{u,v} = \begin{cases} 0 & u = v \\ -\infty & u \neq v \end{cases} \quad u, v \in \mathcal{S}. \quad (2)$$

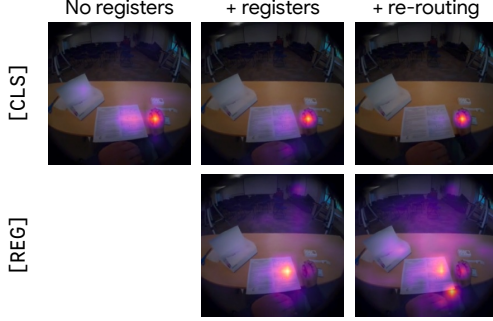


Figure 4: Attention heatmaps for token routing. Adding REG tokens improves action localization in CLS, as non-action content begins to migrate elsewhere. Adding the proposed row-wise token routing sharpens the separation of action content.

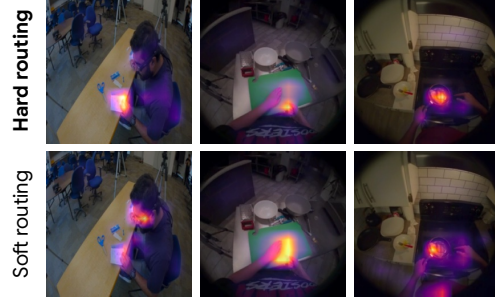


Figure 5: Attention heatmaps for token routing. Keeping the row-wise token routing soft leaks non-action-related content into CLS, whereas our hard routing is more strict to only focus on where dynamics are happening.

Second, each video token is hard-routed to exactly one special token. We compute the mean attention score $\bar{A}$ from video token u to each special token v across heads, and assign the mask in Eq. (3):

$$k_u^* = \arg \max_{v \in \mathcal{S}} \bar{A}_{u,v}, \quad \mathbf{M}_{u,v} = \begin{cases} 0 & v = k_u^* \\ -\infty & v \neq k_u^* \end{cases} \quad u \in \mathcal{X}, v \in \mathcal{S}. \quad (3)$$

The mask is parameter-free. With Eq. (3), each video token must pick a single read source from [CLS, REG] at every layer. Without this restriction, non-action tokens pull in a fraction of CLS’s action-aligned content each layer and start to look action-relevant even when they carry no real signal. Forcing one read source blocks this contamination. Since the action losses in Section 3.2 only shape CLS, its limited softmax bandwidth concentrates on genuinely action-discriminative tokens, while the rest receive vanishing attention and their argmax settles on REG. REGs thus act as sinks for non-action content and are discarded at inference. The diagonal mask in Eq. (2) further prevents CLS from reading REG directly, closing another leak.

Fig. 4 and Table 3 (attention guiding rows) reveal the benefit of our routing. Without the mask, a non-action video token would repeatedly absorb CLS content and then be read back by CLS, giving CLS an indirect path to attend to itself and inflate non-action features into apparent action signal (Fig. 4, first column). Hard-routing such tokens to REG closes this path, so CLS’s effective receptive field is narrowed to video tokens that already carry genuine action signal (Fig. 4, third column).

We also tried column-wise and soft variants, both of which underperformed. Column-wise masking severs the CLS-to-video attention path that carries gradients, while soft masking still diffuses information across special tokens and weakens the separation of action and non-action content (Fig. 5). Related quantitative ablations are reserved to Appendix E. *To our knowledge, no prior work uses routing between special tokens to drive REG into accumulating content away from CLS.*

4 Experiments

In this section, we describe the datasets and training setup, the baselines used, and a broad set of experiments including retrieval, clustering, probing, scene biasing, and a robotic downstream task.

4.1 Setup

Label Definitions. We train using two kinds of labels: atomic sentence-level actions (e.g., *cutting a tomato on a bench*) and atomic verb-level actions (e.g., *cut*). For inference, in addition to atomic ones, we include keystone sentence-level actions (e.g., *check for any damage in the tube*).

Datasets and training. We train and evaluate on Ego-Exo4D [19], using the time-synchronized ego-exo clips with keystone and atomic action labels. Unlike recent cross-view works [31, 45] that report results only on the cooking subset, we evaluate on the full validation split combining {cooking,

Table 1: Video-video retrieval hit-rates of methods in the Ego-Exo4D validation set, for both **atomic** and **keystep action** hit-rates. g-x denote ego-exo, respectively. For keystep actions on our method, we use the atomic action checkpoint without finetuning. The best three results are shown in **bold**, underlined, and in *italic*, respectively. SV is single-view (ego), and MV is multi-view (ego-exo).

Method		Atomic action hit-rates (@10)						Keystep action hit-rates (@10)					
		Cross-scene ($\uparrow$)			Cross-view ($\uparrow$)			Cross-scene ($\uparrow$)			Cross-view ($\uparrow$)		
		g $\rightarrow$ g	x $\rightarrow$ x	avg.	g $\rightarrow$ x	x $\rightarrow$ g	avg.	g $\rightarrow$ g	x $\rightarrow$ x	avg.	g $\rightarrow$ x	x $\rightarrow$ g	avg.
Random		3.04	3.01	3.02	2.15	3.70	2.92	6.23	7.35	6.79	6.59	6.72	6.65
Domain agnostic	CLIP [39]	16.01	15.33	15.67	14.03	16.25	15.14	26.76	20.47	23.61	17.71	17.68	17.70
	DINOv3 [43]	19.08	15.11	17.09	13.60	16.32	14.96	28.50	20.40	24.45	24.43	19.42	21.93
	V-JEPA 2 [4]	17.83	14.05	15.94	15.32	18.40	16.86	33.75	21.55	27.65	14.40	17.45	15.93
	FlowFeat [1]	23.56	24.04	23.80	19.70	23.54	21.62	21.25	15.51	18.38	11.84	12.89	12.37
	LiFT [5]	27.84	23.15	25.50	21.71	27.08	24.39	25.88	19.02	22.45	20.79	16.23	18.51
Robot centric	LAPA [50]	25.24	<u>26.52</u>	25.88	15.96	13.82	14.89	13.58	15.28	14.43	15.09	13.94	14.51
	VILA [22]	19.21	19.51	19.36	22.59	<u>29.51</u>	<u>26.05</u>	33.03	29.06	31.04	12.36	7.22	9.79
Human centric (SV)	TimeSFormer [18]	26.21	20.73	23.47	19.55	23.87	21.71	30.50	19.84	25.17	17.55	15.97	16.76
	LaViLa [51]	28.52	<u>24.25</u>	<u>26.39</u>	<u>23.29</u>	23.84	23.57	41.26	17.68	29.47	20.79	16.43	18.61
	EgoVLP [27]	<u>29.61</u>	21.77	25.69	<u>22.34</u>	23.52	22.93	49.82	26.01	<i>37.91</i>	<i>29.94</i>	22.53	26.23
	EgoVLPv2 [38]	29.35	22.00	25.67	21.88	23.54	23.59	<u>48.11</u>	23.84	35.98	27.55	22.43	24.99
	SUM-L [45]	29.58	23.33	<u>26.46</u>	22.29	23.27	22.78	36.04	14.73	25.39	15.71	9.81	12.76
Human centric (MV)	AE2 [49]	14.88	14.06	14.47	15.47	15.39	15.43	29.35	33.13	31.24	11.22	11.41	11.32
	BYOV [37]	16.84	15.94	16.39	14.57	14.33	14.45	41.06	<u>37.49</u>	<u>39.27</u>	20.47	21.06	20.75
	VI Encoder [19]	19.95	18.64	19.30	19.01	20.40	19.71	19.71	16.17	17.94	16.63	17.22	16.93
	Rosetta [31]	22.38	20.95	21.66	21.66	21.21	21.43	43.82	<u>34.42</u>	<u>39.12</u>	39.94	34.77	37.35
Ours		34.03	31.03	32.53	31.55	32.02	31.79	<u>44.25</u>	38.70	41.47	<u>37.98</u>	<u>32.21</u>	<u>35.09</u>



Figure 6: Comparison of different methods on Ego-Exo4D, on atomic-action cross-scene video-video retrieval task. First row are query videos, and others are top-3 matches. Correct and incorrect hits are visualized with green and red, respectively.



Figure 8: Zero-shot video-video retrieval results on NTU RGB+D dataset. InvAct matches the same-labeled actions under different environments and viewpoints, outside the verb set of train data.

health, bike} subsets using their shared atomic-action labels. For zero-shot evaluation, we employ Charades-Ego [42], SSV2 [17], PhyWorld [23], and NTU RGB+D [29, 41], detailed in Appendix F.3.

Baselines. All baselines are listed in Table 1. We include domain agnostic, robotic-centric, and human-centric single- and multi-view baselines. Details are provided in Appendix F.4.

4.2 Downstream Experiments

To understand **InvAct**’s captured action semantics, we test it under progressively harder downstream setups. These experiments assess fine-grained atomic action retrieval across views and scenes, generalization to unseen data, composition into keysteps without additional training, and using the actions in pretraining of VLAs for robotics manipulation.

Atomic action retrieval. Unlike many cross-view methods that match the same action within the same scene and time, we emphasize action-level matching across both viewpoints and scenes, even when objects, context, and timelines differ. We report retrieval hit-rates following prior work [36, 43, 12, 4], for cross-view and cross-scene verb-level atomic actions in Table 1 and Fig. 6.

InvAct significantly outperforms cross-view methods such as Viewpoint Rosetta, single-view methods such as EgoVLPv2, robotics-focused approaches such as LAPA, and general-purpose extractors such as V-JEPA 2 and FlowFeat. Fig. 6 shows reliable retrieval across all ego-exo query and gallery combinations. Cross-view baselines over-rely on appearance cues such as desks and papers, single-view methods fail under view changes, and general-purpose ones miss the true actions.

Zero-shot retrieval. In line with prior work [51, 31, 38], we define zero-shot as testing on a dataset not used during training. We zero-shot evaluate our embeddings on SSV2, Charades-Ego, and NTU RGB+D. Table 2 and Fig. 8 show that InvAct is superior in most cases, and retrieves the same actions across diverse scenes. In Charades, the actions are mostly scene-based (*reading a book, washing dishes, etc.*), so Rosetta performs slightly better than InvAct. See Appendix A for more results.

Combining atomic actions for keystone retrieval. Each keystone spans a longer time window containing a sequence of atomic actions. We utilize our atomic-action transformer for keystone retrieval without additional training by averaging the atomic embeddings within each keystone segment. As shown in Table 1, this simple aggregation is highly effective, matching methods trained directly on keystone labels such as Viewpoint Rosetta and surpassing all other baselines. Notably, InvAct shows the smallest performance change switching between atomic-keystone retrieval.

Pretraining VLA with actions for robotics manipulation. We further evaluate our embeddings on simulated robotic manipulation tasks. Following LAPA [50], we use latent actions for VLA pretraining and measure task success rates. We pretrain OpenVLA [25] with atomic actions from

Table 2: Zero-shot video-video retrieval action top-10 hit-rates of cross-view methods. Ours can generalize to other datasets better. * are subsets where we omit the training set atomic verb-level actions.

	SSV2	*	Charades	*	NTU	*
VI Encoder	23.68	44.83	15.60	39.34	36.61	25.99
SUM-L	24.92	53.71	13.88	36.21	52.70	65.76
Rosetta	24.96	48.91	17.98	39.91	58.79	73.86
Ours	29.00	55.75	16.45	39.85	60.69	74.23

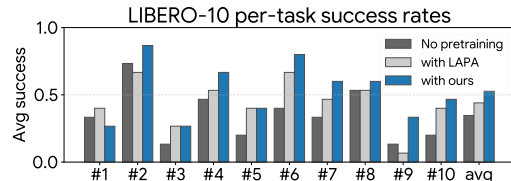


Figure 7: Effect of pretraining with atomic actions on LIBERO-10 tabletop manipulation test suite. Like LAPA, our actions also improve the task success rate.

LAPA (in-domain) and InvAct (zero-shot) on SSV2, then post-train on LIBERO [28] and evaluate on LIBERO-10. As shown in Fig. 7, the baseline achieves 34% success rate, LAPA improves by +10%, and InvAct by +18%, indicating that our representations effectively bridge human- and robot-centric action learning. Details are in Appendix C.

4.3 Understanding the Representation

Beyond downstream performance, we analyze what information InvAct can encode and how it structures action semantics. Specifically, we examine its scene-bias properties, probing consistency over long sequences, clustering behavior, alignment with linguistics, and the attention heatmaps. Extended results are in Appendix A.

Semantic alignment. We assess how well atomic-action embeddings capture linguistic verb semantics by comparing their verb-verb similarity structure using WordNet [34]. We group Ego-Exo4D validation embeddings by ground-truth verb labels, compute verb prototypes, and measure the Spearman correlation between the resulting similarity matrix and the WordNet reference. Fig. 9 shows InvAct achieves the highest alignment, indicating better preservation of relative verb semantics.

Ablation studies. We include an extensive quantitative ablation in Table 3. Detailed explanations are provided in Appendix D.

Scene-bias. We measure scene bias with the same-scene fraction at top- k (SSF@ k), the fraction of queries whose top- k retrieval contains at least one gallery clip from the same scene. SSF counts same-scene clips whether they are correct or incorrect hits, keeping it disentangled from action retrieval performance. To account for dataset distribution, we also compute Prior@ k , the probability of drawing at least one same-scene clip in k uniform random samples, averaged over queries. Table 4 reports both. All baselines sit far above the prior, indicating strong scene bias. InvAct achieves the lowest SSF, with a 30% reduction over Viewpoint Rosetta at top-1, and at top-5 and top-25 falls at or below the prior. Details are provided in Appendix B.

5 Conclusion

We present **InvAct**, the first view- and scene-invariant method for atomic action learning. By our proposed components, InvAct mitigates the view-scene overfitting of prior approaches and achieves competitive atomic and keystone retrieval, zero-shot transfer, and robotic action learning. We see this as a step toward unifying human- and robot-oriented latent action spaces, and expect our structured token commitment approach to extend to other domains.

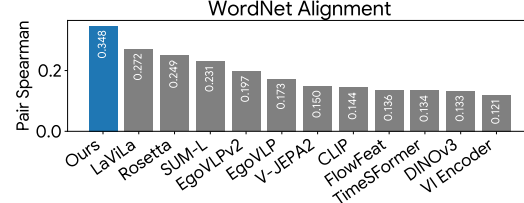


Figure 9: Spearman correlation between each model’s pairwise action-similarity matrix and the WordNet similarity matrix. Our embeddings are more linguistically aligned.

Table 3: Extensive quantitative ablation of InvAct on Ego-Exo4D validation set atomic actions.

	Cross-scene ($\uparrow$)			Cross-view ($\uparrow$)		
	g $\rightarrow$ g	x $\rightarrow$ x	avg.	g $\rightarrow$ x	x $\rightarrow$ g	avg.
<i>Training paradigm</i>						
CLR _{verb}	24.04	23.41	23.72	18.87	22.44	20.65
CE _{verb}	23.70	24.17	23.93	23.47	25.67	24.57
CLR _{atom}	26.82	24.41	25.61	25.99	24.90	25.45
CE _{verb} +CLR _{atom}	27.17	27.75	27.46	27.18	27.74	27.46
<i>Losses</i>						
Text alignment	26.82	24.41	25.61	25.99	24.90	25.45
+View invariance	28.30	24.53	26.41	27.15	27.18	27.16
+Action attract	29.55	24.32	26.93	25.09	25.80	25.45
+Scene repel	29.72	26.67	28.20	27.06	27.23	27.14
<i>Input modality</i>						
RGB	24.44	22.87	23.65	24.20	24.45	24.32
Flow	24.18	23.36	23.77	22.28	23.91	23.05
RGB+flow	25.45	24.46	25.96	24.06	24.88	24.47
CLIP+flow	29.72	26.67	28.20	27.06	27.23	27.14
DINOv3+flow	31.57	28.92	30.25	29.86	29.17	29.52
<i>Attention guiding</i>						
Only CLS token	32.52	30.28	31.40	31.01	30.15	30.58
+REG tokens	32.66	30.59	31.62	31.16	30.91	31.04
+Token re-routing	33.55	31.01	32.28	31.23	31.76	31.50
Trained longer	34.03	31.03	32.53	31.55	32.02	31.79

Table 4: Same-scene fraction at top- k . Lower is better. Prior denotes the dataset-matched reference distribution. InvAct is the least scene-biased method across all top- k values.

	Top-1	Top-5	Top-10	Top-25
Prior@ k	0.39	0.89	0.90	0.95
CLIP	0.99	1.00	1.00	1.00
DINOv3	0.99	0.99	1.00	1.00
V-JEPA 2	0.99	0.99	0.99	0.99
FlowFeat	0.99	1.00	1.00	1.00
LiFT	0.99	0.99	0.99	0.99
LAPA	0.60	0.86	0.99	1.00
TimeSFormer	0.99	0.99	0.99	1.00
LaViLa	0.86	0.96	0.97	0.99
EgoVLP	0.99	0.99	0.99	0.99
EgoVLPv2	0.99	0.99	0.99	0.99
VI Encoder	0.99	0.99	1.00	1.00
SUM-L	0.94	0.98	0.99	0.99
Rosetta	0.99	0.99	1.00	1.00
InvAct	0.52	0.86	0.92	0.93

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Technical Appendices for InvAct

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The Appendices are organized as follows. Section A provides additional qualitative results including zero-shot and in-domain retrieval, action probing, attention map visualizations, and latent clustering. Section B analyzes scene bias across all baselines via same-scene fraction measurements. Section C details the VLA pretraining and finetuning experiments. Section D provides an extensive ablation study comparing loss formulations, input types, and training paradigms. Section E further analyzes the proposed attention re-routing mechanism. Section F covers implementation details including hyperparameters, the full loss formulation, dataset descriptions, and baseline configurations.

A Additional visuals on retrieval, probing, clustering, and attention maps

We provide additional zero-shot retrieval results in Figs. 10 to 12, verb-level atomic action probing in Fig. 13, and in-domain ego-exo retrieval in Figs. 17 and 18. Additional attention map visualizations, t-SNE and PCA clustering results are shown in Figs. 14 to 16.

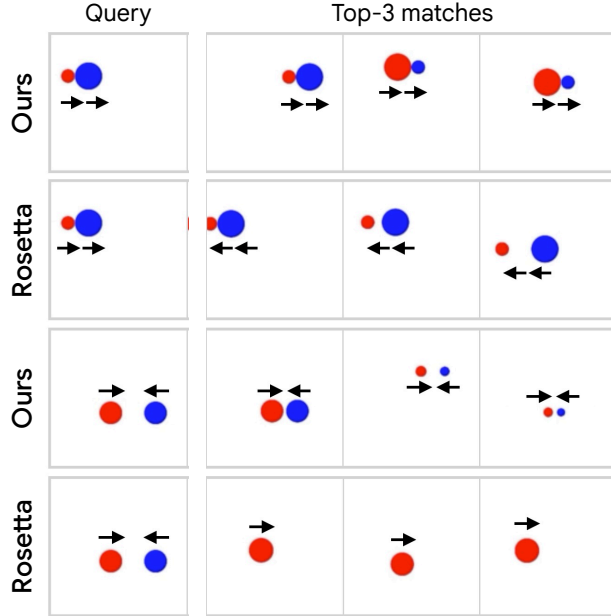


Figure 10: Zero-shot video-video retrieval results on PhyWorld [23]. Arrows denote movement directions in the video (actions). Our method faithfully retrieves correct fine-grained actions, whereas others may opt for color-shape matching (row 2), or retrieve partially-correct actions (row 4).



Figure 11: Zero-shot qualitative video-video retrieval results of our method on SSV2 and Charades-Ego datasets. Our method can successfully extract the actions and populate top-3 matches.



Figure 12: Zero-shot video-video retrieval results on NTU RGB+D dataset of our method. Our action embeddings can match the same-labeled actions, with different environments and camera viewpoints.

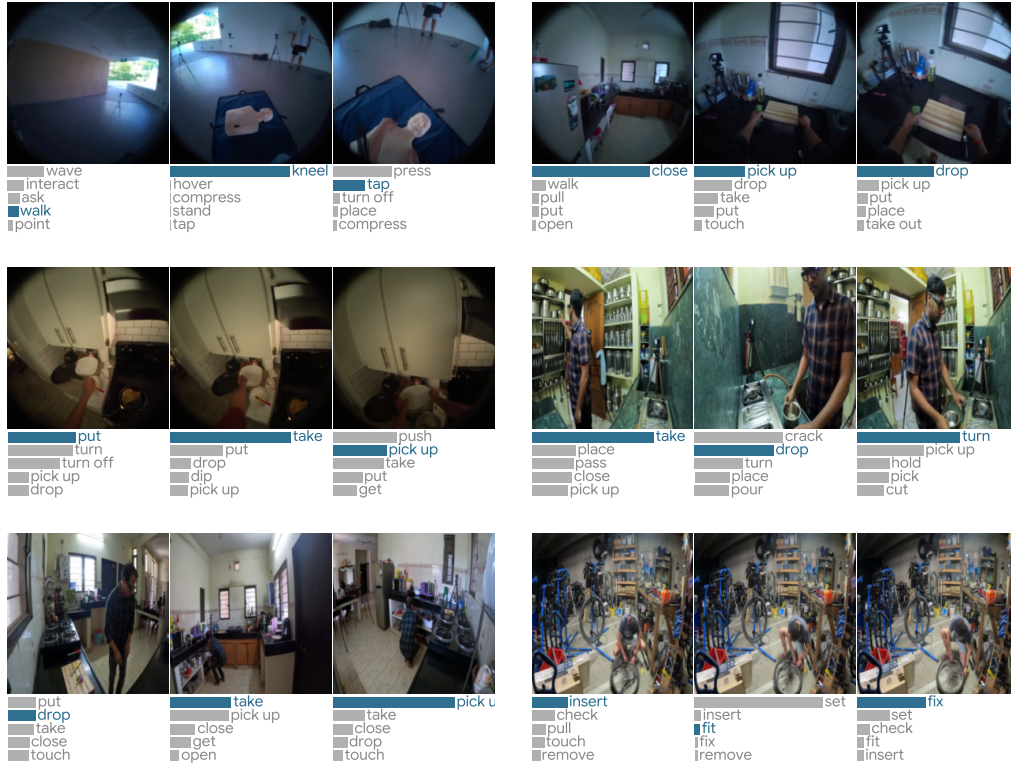


Figure 13: Qualitative atomic action probing results of our method on Ego-Exo4D validation set, for ego and exo videos. Despite not being autoregressive or not having context memory, our method can successfully support longer-range atomic sequences.



Figure 14: Cross-attention heatmaps from our model (bottom row) and the original video frames (top row). Brighter regions are where CLS tokens attend the most. Our action embeddings attend to interaction regions the most.

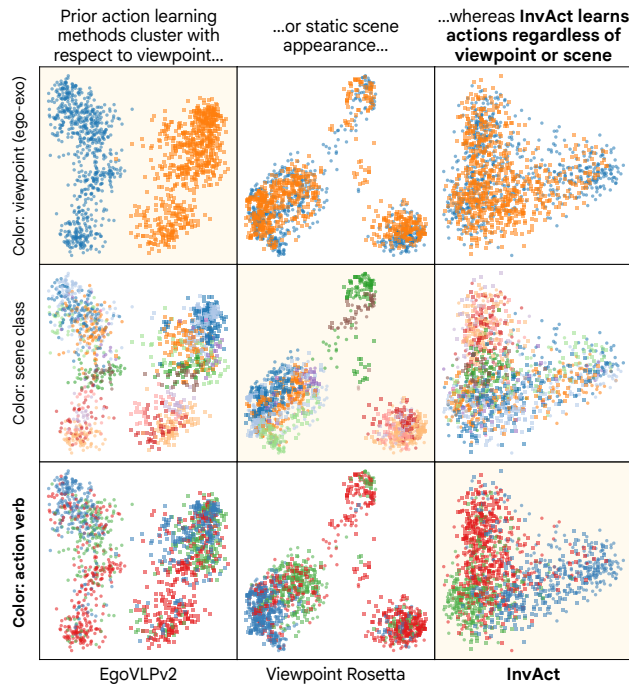


Figure 15: PCA visualization of embeddings from EgoVLPv2, Viewpoint Rosetta, and InvAct. Each column shows the same embeddings under three different colorings (rows). The diagonal highlights the dominant clustering axis for each method. EgoVLPv2 clusters by viewpoint (ego vs. exo), Viewpoint Rosetta clusters by scene class, and InvAct clusters by action type. In the last row, pick, remove, and cut actions are shown in green, red, and blue.

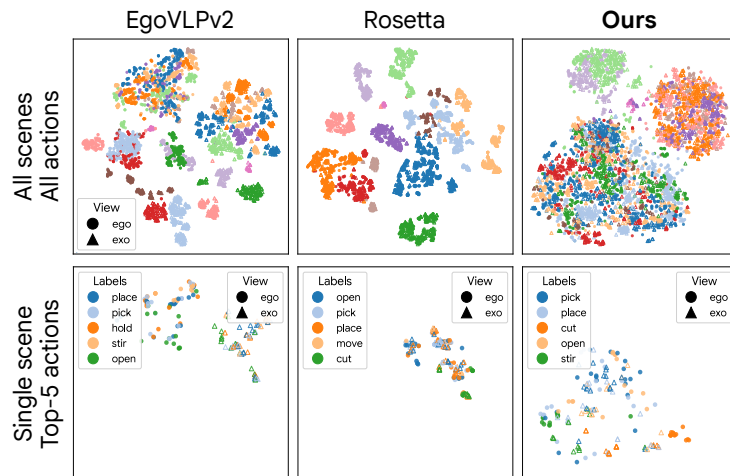


Figure 16: We visualize embeddings with t-SNE in all scenes/all actions (row 1), and a single scene for the top-5 actions (row 2). Different colors denote different scenes in row 1, and different colors denote different verbs in row 2. Prior methods cluster largely by scene or viewpoint, while our embeddings are less scene-dependent and preserve better action grouping across scenes and views.

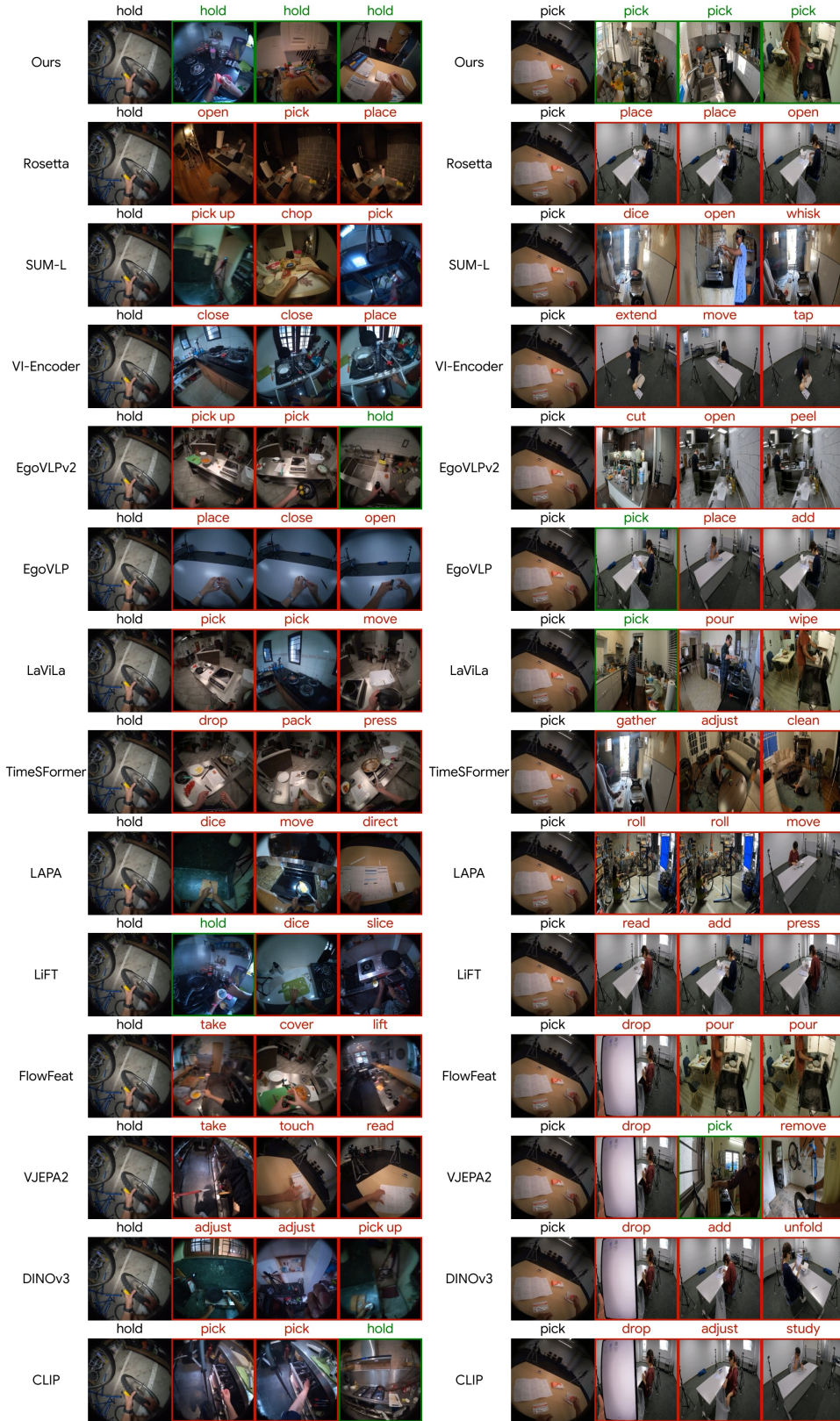


Figure 17: Video-video cross-scene **ego-ego** (left) and **ego-exo** (right) retrieval results on Ego-Exo4D dataset. First columns are the query videos, and other columns are top-3 hits.

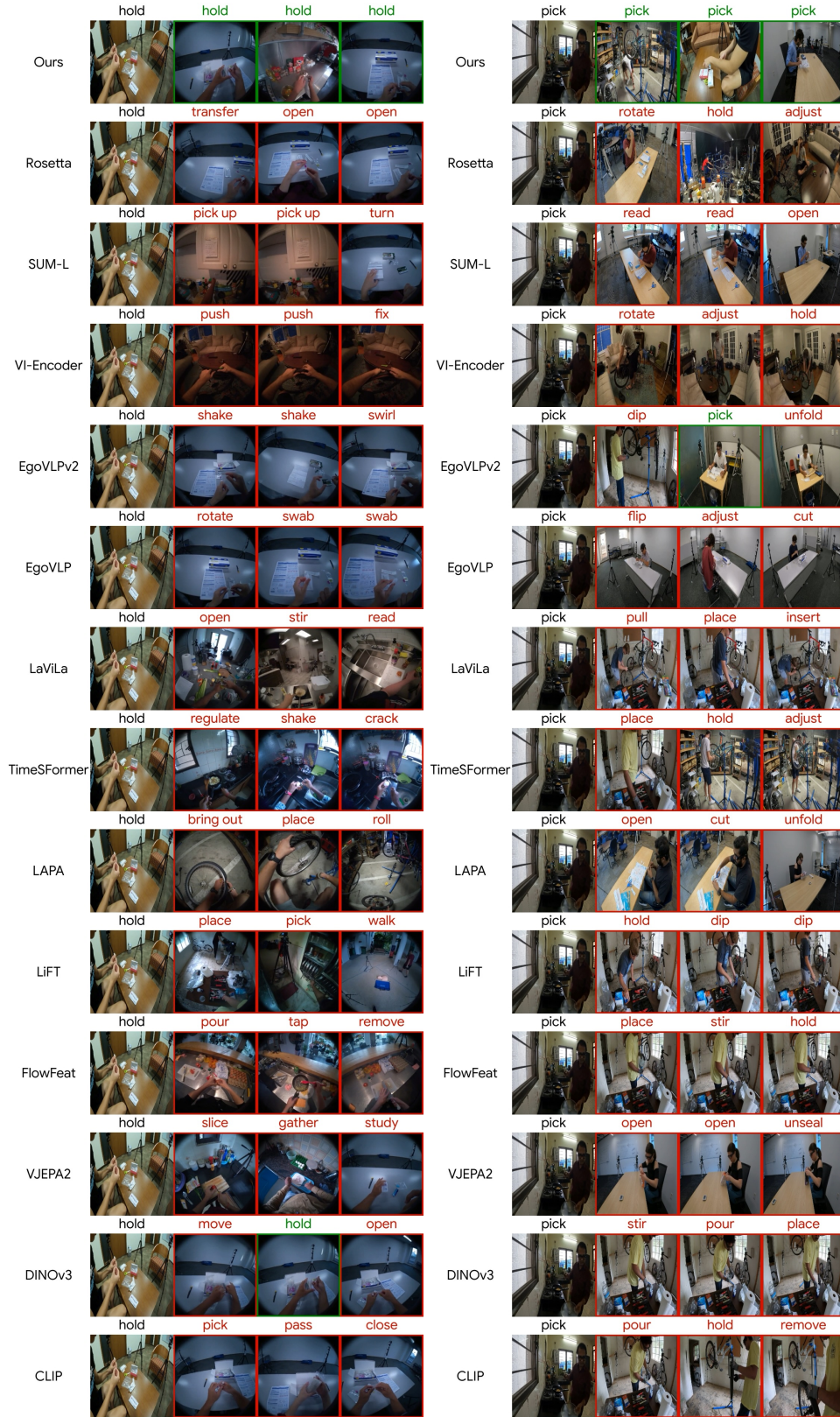


Figure 18: Video-video cross-scene **exo-ego** (left) and **exo-exo** (right) retrieval results on Ego-Exo4D dataset. First column is the query video, and other columns are top-3 hits.

B Measuring scene bias

Same-scene fraction (SSF). To quantitatively measure scene bias, we compute the same-scene fraction at top- k (SSF@ k) as the fraction of queries whose top- k retrieval set contains at least one gallery clip from the same scene as the query:

$$\text{SSF@}k = \frac{\# \text{ queries whose top-}k \text{ contains at least one same-scene clip}}{\# \text{ queries}} \quad (4)$$

Importantly, SSF counts a query as same-scene-biased regardless of whether the retrieved same-scene clip is a positive or negative hit. This keeps SSF disentangled from action retrieval performance and allows it to measure scene bias alone. A lower SSF@ k means fewer queries surface any same-scene clip in their top- k , indicating less reliance on scene cues.

Reference prior. To account for dataset distribution, we compute a retrieval-matched same-scene prior at top- k . For each query q with N_q total gallery clips and M_q same-scene gallery clips, the probability of sampling at least one same-scene clip uniformly without replacement is given in Eq. (5):

$$\text{Prior@}k(q) = 1 - \frac{\binom{N_q - M_q}{k}}{\binom{N_q}{k}}. \quad (5)$$

We average this over all queries to obtain a reference SSF that reflects the gallery distribution alone, independent of any model. Comparing a model’s SSF to this prior reveals whether its retrievals are more or less scene-biased than expected by chance.

Table 5 reports SSF for all methods alongside the dataset prior. All prior methods exhibit SSF values substantially above the prior, indicating strong scene bias. InvAct achieves the lowest SSF across all top- k values, with a 30% reduction over the strongest baseline Viewpoint Rosetta at top-1. Notably, at top-5 and top-25, InvAct falls at or below the dataset prior, meaning our retrievals are no more scene-biased than random sampling from the gallery. This directly validates our proposed scene-aware contrastive objective and token routing mechanism in reducing scene bias without sacrificing action retrieval performance.

Table 5: Same-scene fraction (SSF) at top- k . Lower is better. Prior denotes the dataset-matched reference distribution. InvAct is the least scene-biased method across all top- k values and is the only method to approach or fall below the dataset prior.

Model	Top-1	Top-5	Top-10	Top-25
Data prior	0.39	0.89	0.90	0.95
CLIP	0.99	1.00	1.00	1.00
DINOv3	0.99	0.99	1.00	1.00
V-JEPA 2	0.99	0.99	0.99	0.99
FlowFeat	0.99	1.00	1.00	1.00
LiFT	0.99	0.99	0.99	0.99
LAPA	0.60	0.86	0.99	1.00
TimeSFormer	0.99	0.99	0.99	1.00
LaViLa	0.86	0.96	0.97	0.99
EgoVLP	0.99	0.99	0.99	0.99
EgoVLPv2	0.99	0.99	0.99	0.99
VI Encoder	0.99	0.99	1.00	1.00
SUM-L	0.94	0.98	0.99	0.99
Rosetta	0.99	0.99	1.00	1.00
InvAct	0.52	0.86	0.92	0.93

C VLA pretraining for robotics manipulation task

We follow the same three-stage pipeline as LAPA [50]: (1) learn a latent action space, (2) pretrain a vision–language–action (VLA) model to predict these latent actions, and (3) post-train the VLA to output robot control actions for simulation. We use OpenVLA [25] as the VLA backbone and LIBERO [28] for robot post-training and evaluation.

Stage 1: Latent actions. We obtain two latent spaces: LAPA’s pretrained LAQ model trained on Something–Something V2 (SSV2) [17], and our atomic-action model trained on Ego-Exo4D [19].

Stage 2: VLA pretraining with latents. We finetune OpenVLA using 32-rank adapters [21] with batch size 32 on SSV2. Inputs are video frames and the tokenized action text description (e.g., *rolling something on a flat surface*). We attach a single linear head that maps transformer tokens to the latent action vector and train the VLA to match either the non-quantized LAQ encoder outputs or our latent actions. Both variants are trained for 25,000 steps on a single H200 GPU. The baseline model in Fig. 7 skips the Stage 2 pretraining with SSV2 dataset, and only shows the raw performance, where we focus on comparing LAPA with InvAct.

Stage 3: Robot post-training. We remove the latent-action head, add a new 32-rank LoRA, and finetune on LIBERO to predict continuous robot embodiment actions used by the simulator controller. We add a new trainable output layer that maps transformer tokens to robot actions. In the end, we train three VLA models: (i) no latent pretraining, (ii) latent-pretrained with LAPA, and (iii) latent-pretrained with our embeddings. All use batch size 32 and are trained for 50,000 steps on a single H200 GPU.

Evaluation. We report the average success on the LIBERO-10 test suite with three trained models. For each of the 10 tasks, we run 20 trials and average success rates across tasks. The visuals for the trials per task are given in Fig. 19. Each row in the figures describes a task, totaling up to 10 tasks. The task goals are given in text, and the locations of the objects are randomized per trial.

The task goals are given:

1. put both the alphabet soup and the tomato sauce in the basket
2. put both the cream cheese box and the butter in the basket
3. turn on the stove and put the moka pot on it
4. put the black bowl in the bottom drawer of the cabinet and close it
5. put the white mug on the left plate and put the yellow and white mug on the right plate
6. pick up the book and place it in the back compartment of the caddy
7. put the white mug on the plate and put the chocolate pudding to the right of the plate
8. put both the alphabet soup and the cream cheese box in the basket
9. put both moka pots on the stove
10. put the yellow and white mug in the microwave and close it



Figure 19: Successful (left) and failure (right) runs on LIBERO-10 evaluation suite. Each row is a task. We display 4 evenly subsampled frames from simulation output in each success/failure case.

D Hyperparameter ablation

We perform an extensive ablation with regard to training, input types, losses, and attention guiding in Table 6. We train for fewer iterations for ablation, and train the best model for twice the iterations.

Table 6: Video-video retrieval atomic action hit-rates of our method, in all scenes of Ego-Exo4D validation set. CE and CLR denote cross-entropy and contrastive learning, respectively. Action attract and scene repel construct the unified scene-aware contrastive loss.

	Cross-scene ($\uparrow$)			Cross-view ($\uparrow$)		
	g $\rightarrow$ g	x $\rightarrow$ x	avg.	g $\rightarrow$ x	x $\rightarrow$ g	avg.
<i>Training paradigm</i>						
CLR _{verb}	24.04	23.41	23.72	18.87	22.44	20.65
CE _{verb}	23.70	24.17	23.93	23.47	25.67	24.57
CLR _{atom}	26.82	24.41	25.61	25.99	24.90	25.45
CE _{verb} +CLR _{atom}	27.17	27.75	27.46	27.18	27.74	27.46
<i>Losses</i>						
Text alignment	26.82	24.41	25.61	25.99	24.90	25.45
+View invariance	28.30	24.53	26.41	27.15	27.18	27.16
+Action attract	29.55	24.32	26.93	25.09	25.80	25.45
+Scene repel	29.72	26.67	28.20	27.06	27.23	27.14
<i>Input modality</i>						
RGB	24.44	22.87	23.65	24.20	24.45	24.32
Flow	24.18	23.36	23.77	22.28	23.91	23.05
RGB+flow	25.45	24.46	25.96	24.06	24.88	24.47
CLIP+flow	29.72	26.67	28.20	27.06	27.23	27.14
DINOv3+flow	31.57	28.92	30.25	29.86	29.17	29.52
<i>Attention guiding</i>						
Only CLS token	32.52	30.28	31.40	31.01	30.15	30.58
+REG tokens	32.66	30.59	31.62	31.16	30.91	31.04
+Attn re-routing	33.55	31.01	32.28	31.23	31.76	31.50
Trained longer	34.03	31.03	32.53	31.55	32.02	31.79

Training paradigm. We compare verb-level and sentence-level (atomic) supervision under contrastive (CLR) and cross-entropy (CE) objectives. Verb-level CLR performs poorly on cross-view retrieval, suggesting that coarse verb labels provide an unstable contrastive signal. Switching to verb-level CE substantially improves cross-view performance (+19.0%). Using sentence-level atomic captions with CLR is overall stronger: CLR_{atom} improves the cross-scene and cross-view averages over CE_{verb} by +7.0% and +3.6%, respectively. Finally, combining the two losses CE_{verb} + CLR_{atom} yields the best results, increasing CLR_{atom} by +7.2% (cross-scene) and +7.9% (cross-view), indicating that the two objectives are complementary.

Losses. Starting from text alignment (CLR_{atom}), adding view-invariance improves both cross-scene and cross-view retrieval, with a larger gain on cross-view (+6.7%). Adding action attract further improves the cross-scene (+5.2%), but does not improve the cross-view on average. This is expected since action-attract promotes cross-scene consistency by tightening action clusters, yet without an explicit repel component it can over-collapse representations and disturb viewpoint-specific structure. Finally, adding scene repel recovers and yields the strongest overall results, boosting the cross-scene (+10.1%) and the cross-view averages (+6.6%) over the base case.

Input modality. RGB-only features perform well for cross-view retrieval, but they are largely driven by appearance, which transfers less reliably across scenes. Optical flow provides a complementary motion prior, yielding modest gains in cross-scene retrieval (+2.1% on the hardest split, +0.5% on the cross-scene average). However, flow alone degrades cross-view performance (-5.2%), indicating that motion is informative but insufficient for ego-exo matching. Combining RGB and flow mitigates these failure modes (+9.8% cross-scene avg., +0.6% cross-view avg. over RGB). Using foundation embeddings as guidance further improves: CLIP+flow increases the cross-scene and cross-view averages by +19.2% and +11.6%, and DINOv3+flow achieves the best (+27.9% and +21.4%).

In contrast to approaches that fine-tune pretrained CLIP image encoders as their backbone [31, 49], we only use the output of such pretrained encoders as semantic guidance via cross-attention.

Attention guiding. Starting from the DINOv3+flow baseline (30.25% and 29.52% on cross-scene and cross-view averages respectively), using only a CLS token yields a gain of +1.15% and +1.06%. Adding REG tokens brings a further +0.22% and +0.46%. Introducing our proposed attention re-routing contributes an additional +0.66% and +0.46%, confirming that structurally separating action and non-action content leads to more discriminative representations. Training longer yields the final gains of +2.28% and +2.27% over the DINOv3+flow baseline, establishing our final model at 32.53% and 31.79%.

E Alternative attention re-routing ways

Table 7: Video-video retrieval atomic action hit-rates of our method, in all scenes of Ego-Exo4D validation set. This table is the extension of *attention guiding* experiments in Table 6.

	Cross-scene ($\uparrow$)			Cross-view ($\uparrow$)		
	$g \rightarrow g$	$x \rightarrow x$	avg.	$g \rightarrow x$	$x \rightarrow g$	avg.
Only CLS token = (A)	32.52	30.28	31.40	31.01	30.15	30.58
(A) + REG tokens = (B)	32.66	30.59	31.62	31.16	30.91	31.04
(B) + attn re-routing (column-wise, transpose region)	24.74	23.54	24.14	22.44	23.69	23.06
(B) + attn re-routing (soft)	29.66	27.10	28.38	29.42	30.06	29.74
(B) + attn re-routing (hard, ours)	33.55	31.01	32.28	31.23	31.76	31.50

Column-wise masking of the transposed region vs. row-wise masking (ours). Row-wise masking on the bottom-left sub-matrix constrains each video token, as a query, to read from one special token, while CLS as a query remains free to attend to all video tokens. The action loss therefore receives gradient signal through every video token regardless of routing assignment, and the routing co-adapts so that action-relevant tokens commit to CLS while non-action tokens commit to REG slots.

Column-wise masking on the top-right sub-matrix is the write-side dual: each video token is read by exactly one special token. We conjecture that the critical difference is that under column-wise masking, video tokens routed to REG become directly unreadable by CLS: the attention weight from CLS to those tokens is forced to zero, and gradient from the action loss can no longer reach them through this attention edge. If action-relevant tokens are routed to REG, likely early in training when the argmax is uninformed, CLS cannot easily recover, because the high-bandwidth gradient path is closed. Under row-wise masking, the same misrouting only changes which special token those video tokens read from. That is, the tokens themselves remain directly readable by CLS via the unmasked top-right region, so the loss can correct the routing through ordinary gradient descent.

Consistent with above statement, i.e. due to lack of adequate gradients, we observed severe convergence issues with column-wise masking, evidenced by the lower top- k values in Table 7. We therefore apply row-wise masking on the lower-left sub-matrix.

Soft masking vs. hard masking. In the soft variant, mean attention scores $\bar{A}_{u,v}$ are used as additive logit biases, while in the hard variant (ours), each video token commits to exactly one special token via argmax. As shown in Table 7 and Fig. 5, hard masking consistently outperforms soft masking and produces sharper, more action-focused CLS attention maps. Soft masking still allows information to leak across special tokens, which decreases performance, especially in cross-scene scenarios. This is expected, because diffusive attention maps indicate more scene reliance.

Our hard re-routing details. The mean attention scores $\bar{A}_{u,v}$ are derived from non-normalized query-key logits during the forward pass, and then the re-routing is applied for that layer. We apply hard re-routing on all (3) transformer layers. A total of 3 REG tokens are utilized, and visuals show the averaged heatmaps for all REG. We do not use annealing or warm-up scheduling for the hard mask.

F Training details

F.1 Hyperparameters and batch sampling

We train our method, initialized from random weights, distributed on $4 \times \text{H200}$ GPUs for 50,000 steps, with bf16 mixed precision, using AdamW optimizer and constant learning rate of $1\text{e-}4$.

Each minibatch has 4 ego-exo time-synchronized video tuples, resulting in 8 videos per minibatch. Concretely, each minibatch is a quadruplet containing two actions from the set $\mathcal{A}=\{\text{A1}, \text{A2}\}$ and three scenes from the set $\mathcal{S}=\{\text{S1}, \text{S2}, \text{S3}\}$. We opt for the layout $[\text{S1A1}, \text{S2A1}, \text{S3A1}, \text{S3A2}] \subset \mathcal{S} \times \mathcal{A}$. The sets $\mathcal{A}$ and $\mathcal{S}$ are random subsets of the whole Ego-Exo4D training split for each iteration; therefore, the elements in $\mathcal{A}$ and $\mathcal{S}$ differ in each sampling operation. For such samples with an empty set for $\mathcal{P}(i)$ in Eq. (1), we skip sample i during training.

Cross-attention of pretrained features (keys and values) is injected into all 3 layers of our transformer. Similarly, hard re-routing mechanism is also applied to all 3 layers.

F.2 Losses

Notation. Let $\mathcal{S}$ and $\mathcal{A}$ denote the sets of scene and action labels, respectively. A paired ego-exo clip is associated with a label in the product space $\mathcal{S} \times \mathcal{A}$. We form our minibatches via sampling from this product space. We index paired samples by i and denote their ego and exo embeddings by $g_i \in \mathbb{R}^d$ and $x_i \in \mathbb{R}^d$, respectively, which are the CLS token outputs of the transformer.

Ego-exo view alignment. For each paired sample i , we align the ego and exo embeddings via the cosine distance:

$$\mathcal{L}_{\text{vv}} = \frac{1}{N} \sum_{i=1}^N \left(1 - \langle g_i, x_i \rangle\right) \quad (6)$$

Unified scene-based contrastive loss. For the unified scene-based contrastive loss, we divide it into two main parts and investigate their collective effects on Table 6. *Scene repel* row corresponds to the unified loss, $\mathcal{L}_{\text{scene}}$. The separated definitions are given in Eqs. (7) and (8).

Separated cross-scene action attraction. For each anchor sample i , we form a set of positives $\mathcal{P}(i)$ that share the same action label but come from different scenes, and a set of negatives $\mathcal{N}(i)$ that correspond to different actions. We then apply a multi-positive InfoNCE [35] objective in ego:

$$\mathcal{L}_{\text{att}}^{\text{ego}} = \frac{1}{N} \sum_{i=1}^N -\log \frac{\sum_{p \in \mathcal{P}(i)} \exp \langle g_i, g_p \rangle}{\sum_{k \in \mathcal{P}(i) \cup \mathcal{N}(i)} \exp \langle g_i, g_k \rangle} \quad (7)$$

and define $\mathcal{L}_{\text{att}}^{\text{exo}}$ analogously by replacing g with x . The final attraction loss averages modalities, $\mathcal{L}_{\text{att}} = \frac{1}{2} (\mathcal{L}_{\text{att}}^{\text{ego}} + \mathcal{L}_{\text{att}}^{\text{exo}})$.

Separated same-scene repulsion. For each anchor sample i , we select a negative set $\mathcal{R}(i)$ consisting of clips from the same scene but with a different action label. We penalize their similarity using a 2-way InfoNCE. In ego space:

$$\mathcal{L}_{\text{rep}}^{\text{ego}} = \frac{1}{N} \sum_{i=1}^N -\log \frac{\exp \langle g_i, g_i \rangle}{\exp \langle g_i, g_i \rangle + \sum_{r \in \mathcal{R}(i)} \exp \langle g_i, g_r \rangle} \quad (8)$$

and $\mathcal{L}_{\text{rep}}^{\text{exo}}$ is defined analogously by replacing g with x . Then both modalities are averaged $\mathcal{L}_{\text{rep}} = \frac{1}{2} (\mathcal{L}_{\text{rep}}^{\text{ego}} + \mathcal{L}_{\text{rep}}^{\text{exo}})$.

Atomic sentence-level text-video alignment. Let $t_i^{\text{atom}} \in \mathbb{R}^k$ be the atomic sentence-level pooled CLIP [39] text embedding paired with sample i , and let $\{g_i, x_i\}$ denote the corresponding video embeddings for ego and exo, respectively. We use a symmetric CLIP-style InfoNCE loss with index-matched positives:

$$\mathcal{L}_{\text{text}}^{\text{ego}} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{\exp \langle g_i, t_i^{\text{atom}} \rangle}{\sum_{j=1}^N \exp \langle g_i, t_j^{\text{atom}} \rangle} + \log \frac{\exp \langle g_i, t_i^{\text{atom}} \rangle}{\sum_{j=1}^N \exp \langle g_j, t_i^{\text{atom}} \rangle} \right] \quad (9)$$

and define $\mathcal{L}_{\text{text}}^{\text{exo}}$ by replacing g with x . We then average two losses, $\mathcal{L}_{\text{text}} = \frac{1}{2}(\mathcal{L}_{\text{text}}^{\text{ego}} + \mathcal{L}_{\text{text}}^{\text{exo}})$.

Atomic verb-level text-video alignment. For samples with verb supervision, we predict logits with a shared head $h(\cdot)$ and minimize the cross-entropy over verb classes. Let $y_i \in \{1, \dots, K\}$ be the verb class index (one-hot $\mathbf{y}_i \in \{0, 1\}^K$), and let $\ell_i^g = h(g_i) \in \mathbb{R}^K$, $\ell_i^x = h(x_i) \in \mathbb{R}^K$. Then

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \left[\log \frac{\exp \ell_{i,y_i}^g}{\sum_{c=1}^K \exp \ell_{i,c}^g} + \log \frac{\exp \ell_{i,y_i}^x}{\sum_{c=1}^K \exp \ell_{i,c}^x} \right] \quad (10)$$

The final loss is a weighted sum of all terms. We follow $\lambda_{\text{vv}} = \lambda_{\text{CE}} = 0.5$, and 1.0 for other coefficients.

$$\mathcal{L} = \lambda_{\text{vv}} \mathcal{L}_{\text{vv}} + \lambda_{\text{scene}} \mathcal{L}_{\text{scene}} + \lambda_{\text{text}} \mathcal{L}_{\text{text}} + \lambda_{\text{CE}} \mathcal{L}_{\text{CE}} \quad (11)$$

F.3 Datasets

Ego-Exo4D (in-domain). For our main experiments, we train and evaluate in Ego-Exo4D [19] using the `downscaled_takes/448` subset of time-synchronized ego-exo clips annotated with both keysteps and atomic actions. Keysteps label longer temporal segments that capture the overall activity, and each keystone window typically contains a sequence of finer-grained atomic-action labels. **We train only on atomic-action labels and evaluate both atomic-action and keystone-action labels.** In contrast to previous cross-view works [31, 45] that often report results only on `cooking`, we evaluate the entire Ego-Exo4D validation split spanning `{cooking, health, bike}`, using their set of shared atomic-action labels. Because the atomic-action labels are provided as free-form sentences rather than a fixed taxonomy, we parse them manually to extract verb classes. We also filter out atomic labels containing typographical errors and discard action clips shorter than 1 s and longer than 5 s. After preprocessing, the dataset provides 30,615 training clips and 8,961 validation clips (each paired with an atomic-action label). In the end, the training split we use includes 365 verb classes and 280 keysteps, while the validation split includes 234 verb classes and 270 keysteps. We obey the training and validation splits of the original `downscaled_takes/448`.

PhyWorld (zero-shot). PhyWorld dataset [23] comes with several synthetic subsets (e.g., `collision`, `trajectory`, etc.), and we focus on `collision` for the visual experiments to compare our method and Viewpoint Rosetta. Because everything is rendered in 2D, there is no real “viewpoint” shift like in ego-exo videos. This means that the changes are mostly in appearance and setup. In `collision`, two balls with different colors, sizes and shapes start in different parts of the frame, collide, and then move away, sometimes in different horizontal directions and at different speeds. When we run retrieval on this subset, Viewpoint Rosetta often matches clips by how the balls look (color/size/shape) instead of how they move (direction/speed after the collision). Our method is more driven by the dynamics, therefore indicating better fine-grained atomic-action understanding. This matches what we see on Ego-Exo4D and reinforces our appearance-overfit claim, qualitatively.

Other zero-shot datasets. We evaluated zero-shot video-video retrieval in Charades-Ego [42] (egocentric split, 9,338 samples), NTU RGB+D [41, 29] (fully exocentric, 56,880 samples) and the validation split of Something-Something V2 [17] (predominantly egocentric, 24,777 samples). Since PhyWorld does not provide text labels, we report score-based metrics on the latter three datasets.

All utilized assets are cited and their licenses are respected. Ego-Exo4D uses a custom signed license that allows some commercial/product-development use. SSV2, Charades-Ego, and NTU RGB+D are research-only/non-commercial unless separately permitted. PhyWorld is CC-BY-4.0, allowing use with attribution.

F.4 Baselines

For all baselines utilized, we use their pretrained checkpoints available in their repositories. The details of their structure, frame numbers T , input resolutions, and output dimensions are given in Table 8.

G Broader impacts

InvAct aims to improve video-based action understanding by learning atomic action representations that are robust to viewpoint, scene, and appearance changes. Such representations may benefit

Table 8: Backbone architectures and embedding configurations for all compared methods. We fetch the backbone names from their public code bases.

Method	Backbone	T	Res.	Dim
Ours	Transformer with RoPE + RAFT + DINOv3	8	256 ²	256
Rosetta	CLIP_OPENAI_TIMESFORMER_BASE [39, 8]	4	224 ²	256
BYOV	CLIP_OPENAI_TIMESFORMER_BASE [37, 39]	32/20	224 ²	256
AE2	ResNet50 + 3D Conv + MLP [49]	32	224 ²	256
LaViLa	CLIP_OPENAI_TIMESFORMER_BASE [39, 8]	16	224 ²	256
SUM-L	MultiTaskSlowFast R101 (8×8) [15]	32	280 ²	397
EgoVLPv2	FrozenInTime base_patch16_224 [7]	32	224 ²	256
EgoVLP	FrozenInTime STT base_patch16_224 [7, 8]	16	224 ²	256
TimeSformer	TimeSformer vit_base_patch16_224 [8]	8	224 ²	768
LAPA	LatentActionQuantization (LAQ) [50]	2	256 ²	128
VILA	VILA stage-1 IDM (LAOM [40]-style CNN) [22]	2	64 ²	128
LiFT	dinov2_vits14 [36]	16	224 ²	768
FlowFeat	dinov2_vits14_yt [36]	8	224 ²	768
V-JEPA 2	vjepa2-vitl-fpc16-256-ssv2 [4]	16	256 ²	1024
DINOv3	dinov3-vits16plus [43]	8	224 ²	384
CLIP	clip-vit-base-patch32 [39]	8	224 ²	768

assistive robotics, skill learning, human–robot interaction, extended reality, and video retrieval systems by enabling models to focus more on interaction dynamics than on static background cues. In particular, the method’s ability to transfer from human videos to robotic manipulation suggests potential for reducing the amount of task-specific robot data needed for training.

At the same time, stronger action representations could be misused for surveillance, behavior profiling, or unauthorized activity monitoring. Because the model is trained on video data and may generalize across environments, deployment should require clear consent, privacy-preserving data handling, and careful restrictions in sensitive settings.

H Limitations and future work

Long video clips are currently divided into atomic segments and embedded independently, so there is no enforced temporal consistency between consecutive embeddings sampled from the same long clip. Despite this, action probing remains qualitatively consistent in practice but produces occasional incorrect hits (see Fig. 13). Autoregressive variants are a natural next step for improving probing consistency.

Scaling to unlabeled data, further robotics deployments, and adapting token-level video-to-text decoding [12] are also promising and planned directions.